

Towards Realistic Lifelong Re-identification: Identity Recurrence with Changing Clothes

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Abstract

Existing lifelong person re-identification (Re-ID) methods assume that each identity maintains a relatively stable appearance distribution over time. However, in real-world scenarios, identities often reappear asynchronously with substantial clothing changes, which is not modeled in existing lifelong Re-ID formulations. We therefore study a realistic lifelong cloth-changing Re-ID (LCC) setting, in which identities reappear asynchronously under substantial clothing changes. This setting leads to two core difficulties: the model must acquire new identities while adapting representations of recurring ones, and at the same time remain robust to the substantial representation drift induced by clothing changes over time, which undermines cross-temporal identity consistency. To address these, we develop a framework that disentangles identity-intrinsic representations from clothing-induced biases, enabling identity modeling beyond appearance changes. We further introduce a Dynamic Identity-Anchor Alignment to maintain stable identity anchors under stage-wise distribution shifts. Experiments on the LTCC and PRCC benchmarks demonstrate superior performance and representational stability across multiple learning stages. Our code is available at https://github.com/joyner-7/LCC_ReID.

1. Introduction

Person re-identification (Re-ID) has traditionally been studied under static training settings (Ye et al., 2021), as shown

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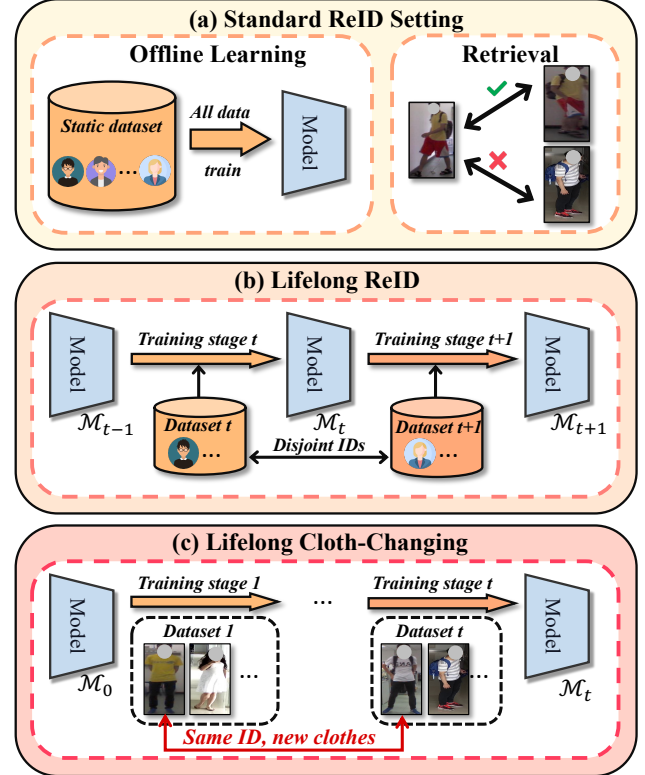


Figure 1. (a) Standard Re-ID uses static offline training. (b) Conventional Lifelong Re-ID assumes disjoint identities across tasks. (c) Our proposed **LCC Re-ID** addresses a realistic scenario where identities recur asynchronously with different clothes.

in Figure 1(a), where models are trained on a fixed and fully accessible dataset (Li et al., 2021a; Ye et al., 2019). However, real-world surveillance systems operate in open and evolving environments, where data arrive as continuous streams over long time horizons. This practical constraint necessitates the shift toward Lifelong Re-ID (LReID), depicted in Figure 1(b), which aims to incrementally incorporate new identities while preserving previously learned knowledge, without suffering from catastrophic forgetting. In realistic long-term surveillance, individuals rarely maintain consistent appearances across time; instead, identities frequently reoccur after extended intervals with substantial changes in clothing and accessories. Such appearance varia-

tions fundamentally violate the assumptions of conventional LReID, where identity representations are implicitly tied to stable visual cues. As a result, effective lifelong adaptation in real-world settings requires a more challenging formulation—Lifelong Cloth-Changing Re-ID (LCC Re-ID) (Figure 1(c))—which explicitly accounts for recurrent identities under drastic appearance changes across learning stages. This setting fundamentally violates the assumptions of appearance consistency and task isolation underlying existing lifelong Re-ID methods. In the LCC setting, each learning stage simultaneously contains genuinely new identities and previously observed ones that reappear under different clothing, yet these two cases demand fundamentally different update behaviors: the former requires effective acquisition of novel identity representations, while the latter calls for adaptation without compromising identity consistency. Moreover, clothing-induced appearance changes introduce substantial representation drift over time, progressively perturbing identity embeddings and weakening cross-temporal associations. Together, these factors make it challenging to continuously integrate new data while preserving stable and discriminative identity representations throughout the lifelong learning process.

To address this challenge, we propose a framework that anchors lifelong learning on identity-intrinsic representations rather than transient appearance cues. The core idea is to decouple identity-related structure from clothing-induced variations, enabling the model to preserve stable identity knowledge while continuously assimilating new observations. Building upon this decoupled representation, we maintain persistent identity anchors that serve as stable references across stages, preventing identity representations from drifting with appearance changes. To further accommodate asynchronous recurrences, we introduce Dynamic Identity-Anchor Alignment, which adaptively reconciles historical anchors with newly observed appearances, ensuring long-term discriminative consistency throughout the lifelong data stream. The main contributions of this work are summarized as follows:

- We formulate a LCC Re-ID setting that departs from conventional identity-disjoint protocols by introducing asynchronous identity recurrence with implicit cross-stage identity overlap, capturing a more realistic long-term surveillance scenario.
- We propose a lifelong learning framework that anchors identity representations on identity-intrinsic cues, effectively decoupling them from transient clothing variations and maintaining consistent identity references under continual appearance evolution via Dynamic Identity-Anchor Alignment.
- We construct an LCC benchmark by reorganizing

Table 1. Comparison of different Re-ID settings. Our LCC Re-ID is the only setting that simultaneously addresses continual learning, appearance changes, and identity recurrence.

Settings	Continual Learning	Cloth Changing	Identity Recurrence
Standard ReID	✗	✗	✗
CC-ReID	✗	✓	✗
Lifelong ReID	✓	✗	✗
LCC ReID	✓	✓	✓

PRCC and LTCC into sequential tasks with cross-stage identity overlap constraints, and demonstrate through extensive experiments that our approach consistently outperforms state-of-the-art lifelong Re-ID methods.

2. Related Work

2.1. Lifelong Person Re-Identification

The paradigm of Lifelong Person Re-Identification (LReID) has emerged to transition Re-ID systems from static training to continuous, open-world adaptation (Pu et al., 2021; Nguyen et al., 2024). The fundamental challenge lies in navigating the stability–plasticity dilemma to prevent catastrophic forgetting (Li & Hoiem, 2017; Kirkpatrick et al., 2017). Early research adapted general continual learning strategies, focusing on weight regularization to preserve critical parameters (Aljundi et al., 2018; Zenke et al., 2017) or rehearsal mechanisms to replay historical distributions (Rebuffi et al., 2017). To circumvent the privacy and storage constraints in exemplar-based systems (Bang et al., 2021), recent methodologies have shifted toward non-exemplar knowledge accumulation via sophisticated prototype modeling (Wu & Gong, 2021; Lu et al., 2025; Xu et al., 2025a;b) and prompt-based tuning. Advanced non-exemplar continual learning methods have also explored feature calibration (Li et al., 2024), distribution-aware prototyping (Xu et al., 2024), prototype reminiscence and knowledge aggregation (Shi & Ye, 2023), and amplitude-aware style replay (Chen et al., 2025) to further preserve historical knowledge while learning new identities. Despite these advances, existing LReID frameworks remain constrained by the assumption of identity-disjoint tasks and short-term appearance consistency (He et al., 2021). Notably, recent efforts such as DKC (Cui et al., 2025) have begun to explore cloth-hybrid lifelong scenarios by incorporating cloth-changing datasets into the training stream. However, these methods typically treat clothing variations as a mere expansion of the data pool within the conventional sequential task protocol, failing to account for the structural evolution of real-world surveillance. We explicitly redefine the experimental formulation to model asynchronous identity recurrence, providing a more suitable benchmark for studying representation drift

caused by cross-stage clothing changes in the same identity.

2.2. Cloth-Changing Person Re-Identification

Cloth-Changing Re-ID (CC-ReID) addresses the retrieval of individuals under substantial appearance variations, thereby relaxing the traditional constant-clothing assumption (Yang et al., 2019). Technical evolution in this domain has diverged into several high-dimensional strategies. Feature disentanglement represents a major trajectory, where adversarial learning or semantic parsing is employed to isolate identity-intrinsic cues from transient appearance attributes (Zhu et al., 2020). Parallel to this, structural modeling leverages auxiliary modalities such as 3D shape or gait to provide invariant biometric constraints (Hong et al., 2021; Jin et al., 2022; Li et al., 2021b). The recent integration of generative AI has further empowered the field, with multi-scale Transformer architectures (Chen et al., 2024) and diffusion-based synthesis significantly enhancing representational stability. Advanced state-of-the-art methods such as AIM (Yang et al., 2023) have pushed the boundaries of static performance. Other recent methods (Liu et al., 2023; Gu et al., 2022; Nguyen et al., 2024; Yang et al., 2022) explore invariant representations via clothing and pose disentanglement, semantic parsing, or gait modeling. However, these methodologies are primarily tailored for offline scenarios where diverse outfits of the same individual are concurrently accessible, making them ill-suited for the continuous and evolving nature of real-world surveillance. This challenge of filtering volatile, non-stationary noise to retain invariant structural relations is also shared by other cross-view matching tasks under severe domain shifts (Zhang et al., 2026). Motivated by this, our framework shifts focus from volatile visual attributes to decoupled, identity-intrinsic representations to handle the evolving lifelong stream.

3. Methodology

3.1. Problem Formulation

We consider Lifelong Cloth-Changing (LCC) Re-Identification over a chronological sequence of K tasks $\mathcal{S} = \{\mathcal{T}_1, \dots, \mathcal{T}_K\}$. Each task $\mathcal{T}_k$ contains a dataset $\mathcal{D}_k = \{(\mathbf{x}_i^k, y_i^k)\}_{i=1}^{N_k}$, where $\mathbf{x}_i^k \in \mathcal{X}$ denotes an input image and $y_i^k \in \mathcal{Y}_k$ its identity label.

Unlike conventional lifelong Re-ID, LCC allows *supervised identity recurrence* across tasks, i.e., $\mathcal{Y}_j \cap \mathcal{Y}_k \neq \emptyset$ for $j \neq k$. To model realistic cloth changes, we impose an *Asynchronous Clothing Constraint*: for any recurring identity, clothing styles in different tasks are non-overlapping. As a result, the conditional appearance distribution is non-stationary across tasks, i.e., $P_j(\mathbf{x} | y) \neq P_k(\mathbf{x} | y)$.

The goal is to learn an embedding function $\Phi : \mathcal{X} \rightarrow \mathbb{R}^d$

that supports discriminative retrieval on the cumulative test set $\mathcal{D}_{\text{test}} = \bigcup_{k=1}^K \mathcal{D}_{\text{test}}^k$. During training on task $\mathcal{T}_k$, only $\mathcal{D}_k$ is accessible, while historical data $\mathcal{D}_{1:k-1}$ are unavailable. The model must embed visually divergent instances of the same identity across tasks and accommodate continuously arriving new identities.

3.2. Framework Overview

To address representation drift and identity recurrence in LCC, we propose a lifelong learning framework that disentangles identity-intrinsic features from transient appearance variations (e.g., clothing and environment). For a batch from task $\mathcal{T}_k$, the backbone extracts intermediate feature maps, which are decomposed into an identity subspace $\mathbf{f}_{\text{id}}$ and a bias subspace $\mathbf{f}_{\text{bias}}$. The identity subspace captures stable, task-invariant cues, while the bias subspace absorbs clothing- and environment-specific factors.

Training follows a two-stage objective. First, a reconstruction decoder preserves information from the decoupled features, aided by a bias-swapping regularization that reinforces the separation of identity and appearance factors. Second, $\mathbf{f}_{\text{id}}$ is optimized with standard metric learning objectives to enforce identity discrimination.

To handle asynchronous identity recurrence, we maintain persistent identity anchors as reference points across tasks. When identities reappear, a selective distillation process aligns current features with their historical anchors, integrating new observations without corrupting long-term identity consistency. We term this mechanism Dynamic Identity-Anchor Alignment.

3.3. Structural Identity Decoupling

In LCC ReID, a central challenge is that the identity-discriminative representation often becomes entangled with stochastic appearance attributes such as clothing styles and illumination. As identities reappear over time with drastically different clothing, this entanglement causes feature drift and unstable identity representations, which hinders long-term recognition (Eom & Ham, 2019).

To address this, we introduce the Structural Identity Decoupling (SID) module, which explicitly separates clothing-invariant identity cues from transient appearance variations. SID operates on intermediate feature tensors $\mathbf{M} \in \mathbb{R}^{C \times H \times W}$ extracted from the backbone, preserving the spatial topology and pedestrian structure. Unlike pixel-level reconstruction that is highly sensitive to colors, textures, or illumination, SID focuses on feature-level reconstruction, retaining high-level structural information while abstracting away nuisance appearance factors.

Concretely, SID projects $\mathbf{M}$ into two disentangled components using independent projection heads ψ_{id} and ψ_{bias} ,

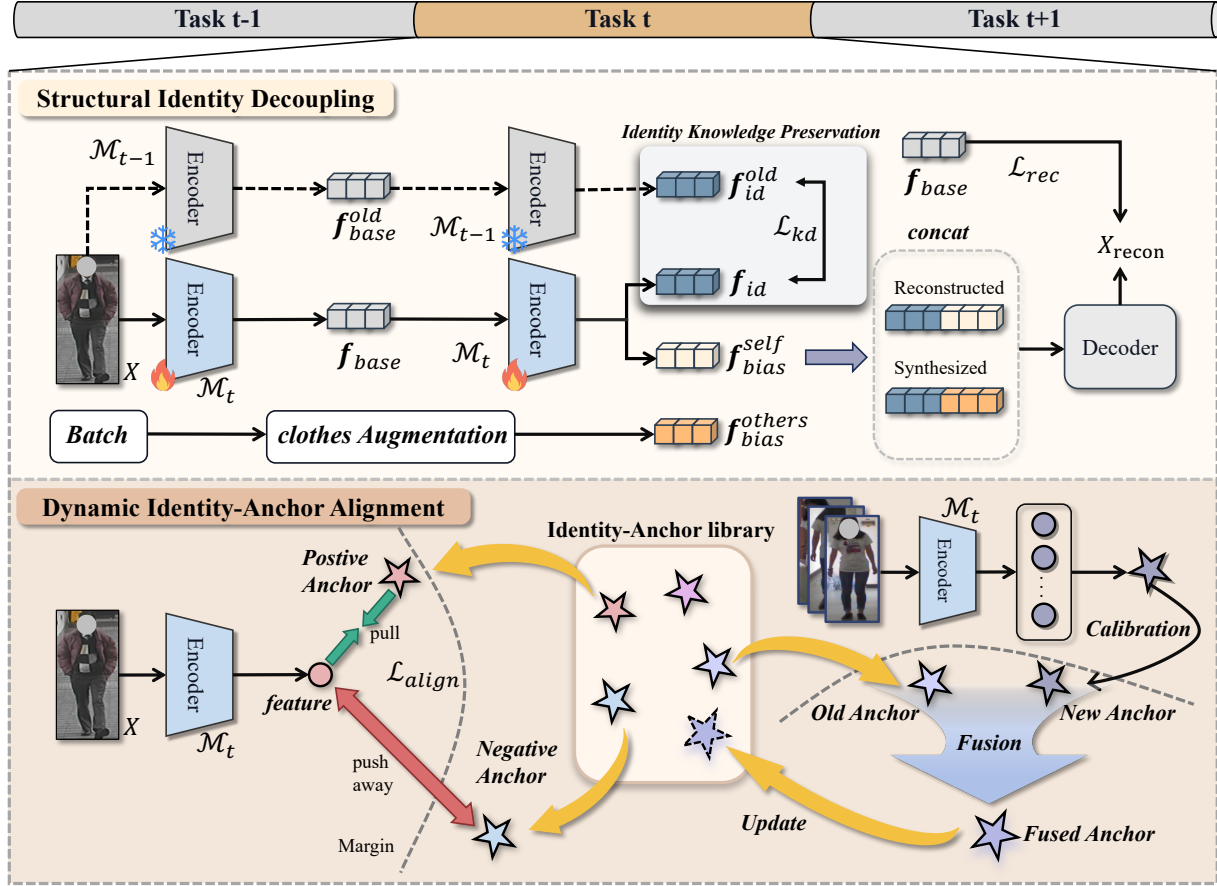


Figure 2. Overview of the proposed framework. The Structural Identity Decoupling (SID) module disentangles clothing-invariant identity features from appearance biases using a bias-swapping mechanism. The Dynamic Identity-Anchor Alignment (DIAA) module maintains long-term consistency by aligning features with a global identity-anchor library and updating anchors via adaptive calibration and fusion.

producing an identity feature $\mathbf{f}_{id} \in \mathbb{R}^d$ and a bias feature $\mathbf{f}_{bias} \in \mathbb{R}^d$. The identity branch captures invariant anatomical and structural cues, while the bias branch absorbs non-stationary appearance attributes such as clothing or lighting variations.

To enforce effective disentanglement, SID employs a Bias-Swapping Reconstruction mechanism. Within a mini-batch, bias features are swapped between samples while keeping identity features fixed, and a shared decoder $D(\cdot)$ is trained to reconstruct the original feature tensor. The reconstruction loss is defined as:

$$\begin{aligned} \mathcal{L}_{rec} = & \|D([\mathbf{f}_{id}, \mathbf{f}_{bias}]) - \mathbf{M}\|_2^2 \\ & + \|D([\mathbf{f}_{id}, \tilde{\mathbf{f}}_{bias}]) - \mathbf{M}\|_2^2, \end{aligned} \quad (1)$$

where $\tilde{\mathbf{f}}_{bias}$ denotes a bias feature sampled from another instance in the mini-batch. By enforcing reconstruction under swapped biases, SID ensures that $\mathbf{f}_{id}$ encodes identity-consistent structural information, while $\mathbf{f}_{bias}$ absorbs clothing-induced variations.

The purified identity representation $\mathbf{f}_{id}$ is further leveraged for Identity-Subspace Distillation (ISD), which explicitly addresses the challenge of recurring identities under clothing changes in LCC ReID. As identities reappear over time with drastically different appearances, maintaining consistent identity embeddings across tasks is critical for long-term recognition. Unlike conventional knowledge distillation that operates on the full feature space and is therefore sensitive to appearance variations, ISD constrains only the identity-intrinsic subspace learned by SID, enabling robust alignment of identity semantics while remaining invariant to clothing changes.

Let $\mathcal{I}_r$ denote the set of identities that have appeared in previous tasks and reoccur in the current task $\mathcal{T}_k$, and let $\mathcal{S}_r = \{i \mid y_i \in \mathcal{I}_r\}$ denote the set of sample indices in the current batch corresponding to these identities. The ISD loss is defined as

$$\mathcal{L}_{isd} = \frac{1}{|\mathcal{S}_r|} \sum_{i \in \mathcal{S}_r} \left\| \mathbf{f}_{id,i}^{(k)} - \text{sg}(\mathbf{f}_{id,i}^{(k-1)}) \right\|_2^2, \quad (2)$$

Algorithm 1 Asynchronous Identity-Anchor Calibration and Update

Input: Existing anchor state $(\mathbf{a}_y, n_y)$, new anchor evidence from current task $(\mathbf{a}_y^{(k)}, n_y^{(k)})$ for identity y

Output: Updated anchor state $(\mathbf{a}_y, n_y)$

// Step 1: Base and query anchor selection based on reliability

if $n_y \geq n_y^{(k)}$ **then**

$(\mathbf{a}_b, n_b) \leftarrow (\mathbf{a}_y, n_y)$

$(\mathbf{a}_q, n_q) \leftarrow (\mathbf{a}_y^{(k)}, n_y^{(k)})$

else

$(\mathbf{a}_b, n_b) \leftarrow (\mathbf{a}_y^{(k)}, n_y^{(k)})$

$(\mathbf{a}_q, n_q) \leftarrow (\mathbf{a}_y, n_y)$

end if

// Step 2: Sample-aware calibration of the query anchor

$\alpha \leftarrow \frac{n_q}{n_b + n_q}$

$\tilde{\mathbf{a}}_q \leftarrow (1 - \alpha)\mathbf{a}_q + \alpha\mathbf{a}_b$

// Step 3: Count-weighted fusion and state update

$\mathbf{a}_y \leftarrow \frac{n_b\mathbf{a}_b + n_q\tilde{\mathbf{a}}_q}{n_b + n_q}$

$n_y \leftarrow n_b + n_q$

where $\mathbf{f}_{\text{id},i}^{(k)}$ and $\mathbf{f}_{\text{id},i}^{(k-1)}$ denote the identity-subspace features of sample i produced by the current and previous models, respectively, and $\text{sg}(\cdot)$ is the stop-gradient operator.

By enforcing consistency of identity-subspace features for recurring identities across successive tasks, ISD preserves long-term identity semantics while remaining invariant to clothing-induced appearance shifts, thereby directly addressing the challenge of asynchronous cloth-changing identity recurrence in lifelong ReID.

3.4. Dynamic Identity-Anchor Alignment

To provide a persistent reference for identity representations under continual appearance evolution, we maintain a set of identity anchors that summarize historical identity knowledge across tasks. Specifically, we construct a Global Identity-Anchor Library $\mathcal{A} = \{(\mathbf{a}_y, n_y)\}_{y \in \mathcal{Y}}$, where $\mathbf{a}_y \in \mathbb{R}^d$ denotes the anchor representation of identity y , and n_y records the cumulative number of observations incorporated into the anchor.

Unlike conventional class centroids aggregated from appearance-dependent features, our identity anchors are defined in the purified identity subspace and serve as stable semantic references across tasks. (Chaudhry et al., 2021) They are not static, but evolve dynamically as new observations of recurring identities arrive over time.

During training on task $\mathcal{T}_k$, we align the current identity representation against the anchor library to preserve global discriminability. Following the implementation, the align-

ment is computed on the L2-normalized identity feature $\mathbf{s}_i$ (obtained by normalizing $\mathbf{f}_{\text{id},i}$). For each sample $\mathbf{s}_i$ with label y_i , we retrieve its positive anchor $\mathbf{a}_{y_i}$ and the nearest negative anchor:

$$\mathbf{a}_{y-} = \arg \min_{y \neq y_i} \|\mathbf{s}_i - \mathbf{a}_y\|_2. \quad (3)$$

We then optimize an anchor-based margin ranking objective:

$$\mathcal{L}_{\text{align}} = [m + \|\mathbf{s}_i - \mathbf{a}_{y_i}\|_2 - \|\mathbf{s}_i - \mathbf{a}_{y-}\|_2]_+, \quad (4)$$

where m denotes a margin hyperparameter. This objective encourages current representations to remain aligned with historical identity anchors while maintaining inter-identity separation across tasks.

A critical challenge in LCC is that identity representations drift over time due to both model updates and the emergence of new clothing styles (Cotogni et al., 2025). When an identity reappears after a long interval, the embedding produced by the updated model may deviate significantly from its historical anchor, even though the underlying identity remains unchanged. Naively aggregating such shifted representations would cause identity anchors to drift, undermining cross-temporal consistency.

To address this, we introduce an asynchronous identity-anchor update mechanism that calibrates incoming anchor evidence before integration, explicitly compensating for model-induced and appearance-induced representation drift.

Given identity y , let $(\mathbf{a}_y, n_y)$ be its existing anchor state, and $(\mathbf{a}_y^{(k)}, n_y^{(k)})$ denote the new anchor evidence extracted from the current task, where $\mathbf{a}_y, \mathbf{a}_y^{(k)} \in \mathbb{R}^d$ and $n_y, n_y^{(k)}$ record the corresponding observation counts. We first choose a base anchor and a query anchor according to their statistical reliability:

$$(\mathbf{a}_b, n_b), (\mathbf{a}_q, n_q) = \begin{cases} (\mathbf{a}_y, n_y), (\mathbf{a}_y^{(k)}, n_y^{(k)}), & n_y \geq n_y^{(k)} \\ (\mathbf{a}_y^{(k)}, n_y^{(k)}), (\mathbf{a}_y, n_y), & n_y < n_y^{(k)}. \end{cases} \quad (5)$$

Here, $(\mathbf{a}_b, n_b)$ denotes the *base anchor* with stronger historical support, while $(\mathbf{a}_q, n_q)$ denotes the *query anchor* extracted under the current model.

To compensate for representation drift between models, we perform a sample-aware calibration that softly rectifies the query anchor toward the base anchor:

$$\alpha = \frac{n_q}{n_b + n_q}, \quad \tilde{\mathbf{a}}_q = (1 - \alpha)\mathbf{a}_q + \alpha\mathbf{a}_b. \quad (6)$$

Finally, the identity anchor is updated via count-weighted fusion:

$$\mathbf{a}_y \leftarrow \frac{n_b\mathbf{a}_b + n_q\tilde{\mathbf{a}}_q}{n_b + n_q}, \quad n_y \leftarrow n_b + n_q. \quad (7)$$

Table 2. Performance comparison of different methods on LTCC and PRCC datasets under same-clothing (SC) and cloth-change (CC) settings. The best results are highlighted in bold.

Methods		LTCC				PRCC			
		same-cloth		cloth-change		same-cloth		cloth-change	
		mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
joint train		44.8	77.3	32.4	58.6	99.9	98.7	63.9	60.1
CIL	LwF(ECCV2018)	20.1	30.1	10.2	15.6	84.2	90.8	17.9	24.3
	PRAKA(ICC2023)	17.2	29.2	8.1	15.2	86.2	93.1	20.2	27.2
	FCS(CVPR2024)	18.1	28.9	9.2	14.8	87.6	92.4	21.1	24.2
CCReID	CAL(CVPR2022)	18.5	38.0	6.7	14.6	96.6	99.5	22.5	28.4
	AIM (CVPR23)	17.6	34.4	7.2	12.5	90.6	95.4	22.1	24.6
	CaAug(MM2024)	21.6	54.3	4.7	12.9	96.6	99.5	33.7	29.5
LReID	AKA(CVPR2021)	16.2	35.4	12.4	21.2	93.1	96.4	24.2	28.2
	DKC(CVPR2025)	15.7	40.4	14.2	24.4	80.2	94.5	21.0	29.0
	DKP(CVPR2024)	9.6	32.9	18.5	33.8	88.5	96.9	27.2	34.6
	DRE(TNNLS2025)	16.8	35.4	14.2	26.8	95.8	95.6	22.8	20.9
	DASK(AAAI2025)	20.2	42.4	17.8	30.4	96.6	97.9	30.1	29.8
	PKA(IJCAI2025)	15.5	39.5	17.2	34.2	93.4	96.2	32.4	30.3
	ADSR(MM2025)	17.1	40.3	15.0	30.9	95.5	97.0	30.0	26.6
Ours		29.4	58	25.8	39.7	96.9	98.4	38.2	37.2

This two-stage update compensates for model-induced and appearance-induced embedding drift, enabling identity anchors to evolve smoothly while preserving long-term identity consistency. The detailed procedure of this calibration and update process is outlined in Algorithm 1.

3.5. Joint Optimization Objective

Our framework is optimized with a unified multi-objective loss that simultaneously addresses the main challenges of lifelong cloth-changing Re-ID. The identity-invariant feature $\mathbf{f}_{id}$ obtained from Structural Identity Decoupling is supervised with a standard Re-ID objective, combining a label-smoothed cross-entropy loss and a hard-mining triplet loss:

$$\mathcal{L}_{reid} = \mathcal{L}_{ce} + \mathcal{L}_{tri}. \quad (8)$$

To further enhance robustness under continual appearance evolution, we incorporate three auxiliary objectives. The reconstruction loss $\mathcal{L}_{rec}$ encourages structural-regularized feature decoupling, helping the model to separate identity-intrinsic cues from transient clothing variations and thus maintain stable identity representations. The alignment loss $\mathcal{L}_{align}$ ensures global discriminability by calibrating the current identity features against the dynamic identity-anchor library maintained by DIAA. Finally, the Identity-Subspace Distillation loss $\mathcal{L}_{isd}$ preserves long-term identity knowledge for recurring individuals by enforcing consistency in the purified identity subspace.

The overall training objective combines these terms in a weighted sum:

$$\mathcal{L}_{total} = \mathcal{L}_{reid} + \lambda_1 \mathcal{L}_{rec} + \lambda_2 \mathcal{L}_{align} + \lambda_3 \mathcal{L}_{isd}, \quad (9)$$

where λ_1 , λ_2 , and λ_3 control the relative contributions of identity discrimination, appearance-invariant representation learning, and long-term knowledge preservation, respectively. By jointly optimizing all components, the model can learn discriminative identity features, resist clothing-induced drift, and retain knowledge from previously observed identities throughout the lifelong learning process.

4. Experiments

4.1. The LCC Benchmark

Unlike conventional lifelong learning protocols (Pu et al., 2021; Wu & Gong, 2021), which assume identity-disjoint task sequences, the LCC ReID benchmark is designed to reflect a common phenomenon in long-term surveillance: the asynchronous evolution of individual appearance. Under this setting, the same identity may reappear across tasks in different outfits, without simultaneous access to multiple clothing variations. This asynchrony introduces the challenge of identity recurrence and cross-stage association, as models must recognize recurring individuals despite substantial visual changes. The resulting variability in appearance can significantly increase the risk of representation drift, making the maintenance of long-term discriminative consistency across sequential tasks more difficult.

4.2. Dataset Specifications

PRCC-LCC. Constructed from the PRCC dataset (Yang et al., 2019), this benchmark comprises 150 identities and 17,896 images, partitioned into 5 sequential tasks. Its pri-

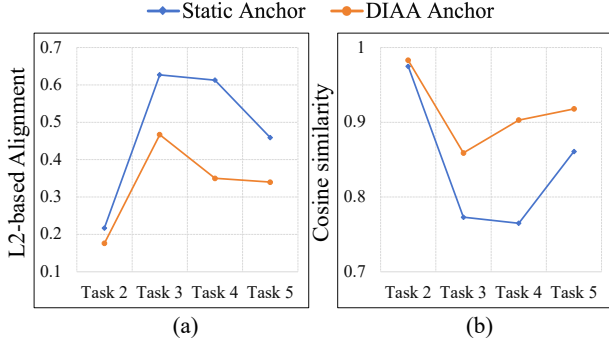


Figure 3. Stability analysis of identity anchors on the LTCC cloth-changing subset. (a) L_2 -based alignment and (b) cosine similarity between current anchors and drift-free reference anchors across sequential tasks. Higher values indicate closer proximity to the ideal identity-intrinsic representations.

mary difficulty stems from the high recurrence rate (78.0%). In this setting, approximately four-fifths of the individuals reappear in subsequent tasks with entirely different outfits after their initial observation. This high-overlap but low-appearance-consistency distribution forces the model to continuously refine the decision boundaries for specific identities, capturing robust anatomical or structural cues amidst severe pixel-level fluctuations.

LTCC-LCC. The LTCC-based benchmark (62 IDs and 5,396 images) (Qian et al., 2020) focuses on simulating extreme and irregular clothing diversity. Each identity is associated with 1 to 5 different outfits, and the recurrence patterns are non-stationary (e.g., some IDs reappear in 4 out of 5 tasks, while others appear only once). This highly imbalanced sampling frequency mimics the uncertainty of human activity in the real world, testing the generalization limits of the disentanglement module and the robustness of the prototype update mechanism when facing infrequent or frequently changing appearances. Task-level statistics are detailed in Table 3.

4.3. Evaluation Protocol and Metrics

We employ the standard metrics in person Re-ID: Rank-1 Cumulative Matching Characteristics and mean Average Precision (mAP) (Zheng et al., 2015). To comprehensively assess the model’s knowledge accumulation and anti-forgetting capabilities, we implement a Global Evaluation Protocol. Specifically, upon completion of each task T_t , the model is evaluated on a fixed, comprehensive test set (10,800 images for PRCC and 7,050 for LTCC) containing all identities from the entire stream. This protocol not only measures the discriminative precision for new outfits (plasticity) but also strictly evaluates the success rate of cross-temporal associations for identities that have undergone significant appearance shifts across different stages

(stability).

4.4. Comparison with State-of-the-Art Methods

To validate the effectiveness of our proposed framework, we conduct a comprehensive comparison against 11 state-of-the-art methods spanning three categories: Class-Incremental Learning (CIL), Cloth-Changing Re-ID (CC-ReID), and Lifelong Re-ID (LReID). All methods are re-implemented on the LCC benchmark to ensure a fair comparison. Quantitative results on the LTCC and PRCC datasets are reported in Table 2.

Comparison with CIL Methods. class incremental learning approaches, including LwF (Li & Hoiem, 2017), PRAKA (Shi & Ye, 2023), and FCS (Li et al., 2024), show limited performance in the LCC setting. Although they leverage regularization or distillation to alleviate forgetting, they are not designed to handle fine-grained open-set retrieval tasks with substantial intra-identity variation caused by clothing changes. For example, FCS achieves only 9.2% mAP on the LTCC cloth-change subset (Table 2), indicating that conventional knowledge-preservation constraints are insufficient for maintaining discriminative identity representations under the dual challenges of catastrophic forgetting and appearance drift.

Comparison with CC-ReID Methods. Specialized Cloth-Changing Re-ID approaches, such as CAL (Gu et al., 2022), AIM (Yang et al., 2023), and CaAug (Liu et al., 2024), are originally optimized for static, offline scenarios where multiple outfits per identity are available simultaneously. When applied to the LCC setting, their performance drops drastically. For instance, CaAug achieves 96.6% mAP on PRCC-SC but only 4.7% mAP on the LTCC cloth-change subset. This decline demonstrates that without explicit lifelong learning mechanisms, CC-ReID models overfit to the clothing distribution of the current task and fail to associate recurring identities appearing in novel outfits across sequential tasks.

Comparison with LReID Methods. State-of-the-art lifelong Re-ID methods (AKA (Pu et al., 2021), DKC (Cui et al., 2025), DKP (Xu et al., 2024), PKA (Lu et al., 2025), ADSR (Chen et al., 2025)), DRE (Liu et al., 2025), DASK (Xu et al., 2025a) effectively navigate the stability-plasticity dilemma but typically rely on the assumption of temporal appearance consistency. The LCC benchmark challenges this premise by introducing significant asynchronous clothing-induced variations. For instance, although DKP is the most competitive LReID baseline, our framework outperforms it by +7.3% mAP and +5.9% Rank-1 on the LTCC cloth-change subset. This performance gap underscores the limitations of conventional rehearsal and distillation mechanisms, which often fail to associate recurrent identities across divergent outfits or incur negative transfer.

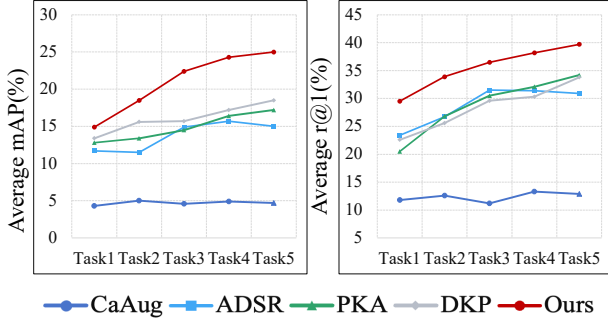


Figure 4. Performance tendency on the LTCC cloth-changing subset across five sequential tasks. After each training stage is completed, the average test performance (mAP and Rank-1) on the cumulative test set is calculated to assess the model’s anti-forgetting capability.

Table 3. Statistical summary of the LCC benchmark. IDs / Outfits denote the count per task.

Task	1	2	3	4	5
PRCC (IDs/Outfits)	53 / 58	53 / 58	53 / 58	57 / 63	51 / 61
LTCC (IDs/Outfits)	22 / 24	20 / 26	24 / 26	22 / 26	21 / 25

4.5. Performance Trends

We analyze the evolution of identity representations under the lifelong cloth-changing setting by examining mAP and Rank-1 scores on the LTCC cloth-change subset across five sequential tasks (Figure 4). The results reveal systematic differences among baselines and our method. Methods designed for static cloth-changing Re-ID, such as CaAug, exhibit minimal improvement across tasks, with mAP consistently between 4.3% and 5% and Rank-1 between 11.2% and 13.3%. This behavior indicates that these methods overfit to the clothing distribution of the current task and cannot associate recurring identities under novel appearances. Lifelong Re-ID baselines, including ADSR and PKA. Despite mitigating catastrophic forgetting, these methods treat drastic clothing variations as uninformative noise, limiting incremental refinement of identity representations.

In contrast, our framework exhibits a robust upward trajectory, with mAP advancing from 14.9% to 25.8% and Rank-1 from 29.5% to 39.7%. This expanding performance margin demonstrates that the synergy between SID and DIAA allows the model to effectively exploit asynchronous identity recurrence. By decoupling identity-intrinsic features from transient appearance, the model interprets new outfits as informative signals for representation consolidation. These results validate that our approach achieves genuine cumulative knowledge integration rather than mere preservation of past states.

Table 4. Analysis of distillation targets. By distilling f_{id} , our method achieves the best stability across both datasets, whereas entangled features (f_{full}) and bias features (f_{bias}) degrade performance.

Dataset	Target	Same-Cloth		Cloth-Change	
		mAP	R-1	mAP	R-1
LTCC	Full (f_{full})	28.2	55.7	23.9	38.0
	Bias (f_{bias})	27.8	56.0	23.8	37.5
	Ours (f_{id})	29.4	58.0	25.8	39.7
PRCC	Full (f_{full})	96.6	98.3	37.6	36.5
	Bias (f_{bias})	96.4	98.4	36.8	35.4
	Ours (f_{id})	96.9	98.4	38.2	37.2

Table 5. Ablation study of SID and DIAA on LTCC and PRCC under same-clothing (SC) and cloth-change (CC) settings.

Methods	SC mAP	SC R@1	CC mAP	CC R@1
LTCC Dataset				
Baseline	18.3	46.3	16.3	32.1
+ DIAA	27.9	56.7	23.8	37.8
+ SID	22.7	52.3	19.9	36.7
Ours (Full)	29.4	58.0	25.8	39.7
PRCC Dataset				
Baseline	95.8	96.5	30.4	27.5
+ DIAA	96.5	99.4	37.8	36.2
+ SID	97.2	98.2	32.3	29.7
Ours (Full)	96.9	98.4	38.2	37.2

4.6. Anchor Update Analysis

We evaluate the effect of the anchor update mechanism in Dynamic Identity-Anchor Alignment (DIAA) by comparing identity anchors with and without updates across sequential tasks in the LCC Re-ID setting. In this context, Static Anchor refers to anchors that remain unchanged, while DIAA Anchor refers to anchors updated through the proposed method. The alignment between identity anchors and reference anchors is quantified using two complementary metrics: L2-based alignment and cosine similarity.

As shown in Figures 3, updating anchors consistently improves both L2-based alignment and cosine similarity across all tasks. This trend becomes more pronounced in later stages, where recurring identities undergo larger clothing-induced appearance variations. Without updates, static anchors gradually drift away from the identity-intrinsic reference anchors due to the accumulation of transient appearance changes, leading to degraded alignment. In contrast, DIAA effectively integrates new observations while suppressing clothing-specific perturbations, preserving the identity-invariant structure of the anchors over time. As a result, identity anchors remain stable and discriminative throughout the lifelong learning process, enabling reliable identity association under realistic cloth-changing scenarios.

We evaluate the individual contributions of Selective Identity Distillation (SID) and Dynamic Identity-Anchor Alignment (DIAA) on the LCC benchmark in Table 5. Incorporating SID into the baseline improves LTCC cloth-changing mAP from 16.3% to 19.9%, confirming that structurally decoupling identity cues from clothing variations provides a robust representation foundation. DIAA alone achieves 23.8% mAP, showing that aligning recurring identities with evolving anchors is critical for mitigating asynchronous drift. The joint deployment of both modules further boosts performance to 25.8% mAP, indicating that the purified identity features and adaptive alignment mechanism are mutually complementary.

4.7. Hyperparameter Sensitivity Analysis

We investigate the sensitivity of four key hyperparameters on the LTCC-LCC benchmark: the alignment margin m , reconstruction weight λ_1 , anchor-alignment weight λ_2 , and ISD weight λ_3 . As illustrated in Figure 5, the proposed framework maintains stable performance across a broad range of settings under both same-clothing (SC) and cloth-change (CC) scenarios, demonstrating strong robustness to parameter variations.

Based on the overall trade-off, we adopt a default configuration of $m = 0.3$, $\lambda_1 = 1.2$, $\lambda_2 = 1.5$, and $\lambda_3 = 0.5$ to achieve balanced mAP and Rank-1 metrics. Specifically, a moderate margin m enhances inter-class discrimination without over-constraining the feature space. Meanwhile, appropriate scaling of λ_1 , λ_2 , and λ_3 provides effective structural decoupling, anchor alignment, and cross-task regularization while preserving the primary Re-ID objective. The smooth performance trends across these settings verify the stable optimization behavior of our framework.

4.8. Rationality of Distillation Targets

To further verify the rationality and effectiveness of our Identity-Subspace Distillation (ISD), we investigate the impact of performing knowledge distillation on different feature components: the complete feature representation ($\mathbf{f}_{full}$), the bias feature ($\mathbf{f}_{bias}$), and our decoupled identity feature ($\mathbf{f}_{id}$). As reported in Table 4, distilling $\mathbf{f}_{full}$ yields suboptimal performance, particularly on the challenging cloth-changing subsets (e.g., a drop of 1.9% mAP on LTCC-CC compared to ours). This degradation suggests that the complete representation entangles identity cues with transient clothing details; thus, enforcing consistency on $\mathbf{f}_{full}$ implicitly suppresses legitimate feature variations required for adapting to new outfits. Furthermore, distilling $\mathbf{f}_{bias}$ results in the lowest performance, confirming that retaining clothing-specific styles introduces noise that disrupts long-term identity association. In contrast, our approach achieves the best results by explicitly isolating the identity-intrinsic

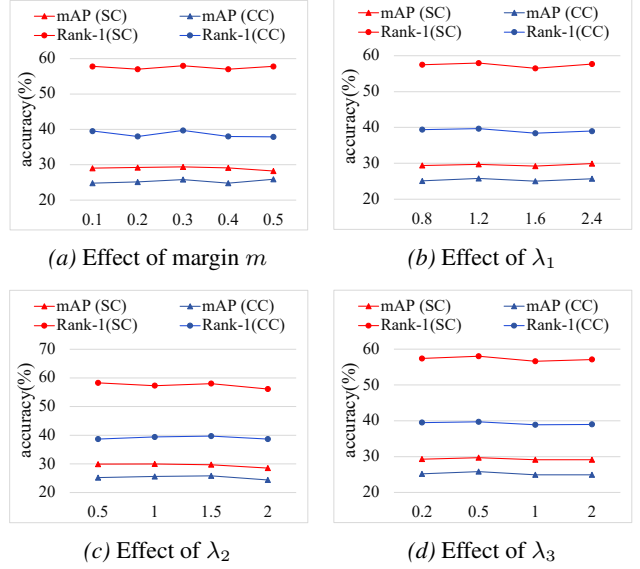


Figure 5. Sensitivity analysis of four critical hyperparameters on the LTCC dataset under same-clothing (SC) and cloth-change (CC) settings.

manifold. By filtering out appearance perturbations before distillation, our method effectively consolidates structural knowledge without overfitting to the specific clothing distributions of historical tasks.

5. Conclusion

We introduce Lifelong Cloth-Changing (LCC) Re-ID, a realistic setting addressing asynchronous identity recurrence under appearance shifts. To tackle representation drift, we propose a framework that decouples identity-intrinsic cues via Structural Identity Decoupling (SID) and maintains long-term consistency through Dynamic Identity-Anchor Alignment (DIAA). Experiments on our constructed benchmarks confirm that our method effectively mitigates catastrophic forgetting and clothing-induced bias, significantly outperforming state-of-the-art baselines.

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Impact Statement

This paper advances person re-identification under long-term appearance changes, benefiting public safety and missing person searches. However, persistent tracking across extended intervals poses privacy risks regarding unauthorized surveillance. To mitigate these dual-use concerns, deployments should be confined to regulated environments with strict data retention limits and must comply with regional biometric data privacy laws to ensure rigorous auditing and accountability.

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